Process Notes: Raw Ratios to Interpretable Cox Graphs

These notes outline a practical workflow for turning raw performance metrics (ratios, pool means, pool ranks) into interpretable Cox plots and model inputs. The goal is consistent interpretation across runs while preserving flexibility.

1) Start With Raw Metrics (0–1 scale)

Raw values preserve immediate meaning: - `tb_ratio_*` and `pool_tb_ratio_mean_*` are ratios from 0–1. - Pool ranks/percentiles are also on 0–1. - Pool differences live around 0 if centered.

Use raw when: - You want the x-axis to be directly interpretable (e.g., 0.20 = 20%). - You want stable meaning across models and papers.

2) Choose the Pool Signal You Want

These are different constructs: - **Pool mean**: "How strong is the pool overall?" (context). - **Pool minus mean (exclude-self mean)**: "What is the pool average without me?" (context). - **Pool diff (individual minus mean)**: "How far above/below the pool am I?" (signal). - **Pool rank / percentile**: "Where do I stand in the pool?" (ordering).

Recommendation: - Use **pool mean** for context. - Use **pool diff** for relative performance. - Use **rank/pct** when you want monotonic, non-parametric ordering.

3) Quadratic Terms: Square Raw First

Quadratic effects should be built from raw values: 1. `x_raw` (ratio or centered diff) 2. `x_sq = x_raw ** 2` 3. Optionally standardize `x_sq` (if needed for model stability)

Why: - Squaring raw preserves interpretability (0–1 stays 0–1). - Squaring after z-scoring creates harder-to-interpret magnitudes.

4) Standardization: When and Why

Use **raw** when: - You want the x-axis to carry direct meaning. - You want to compare across studies or contexts without rescaling.

Use **z-scores** when: - You want coefficient comparability across variables. - You are combining variables in interactions with different scales.

Important rule: **Raw → square → then (optional) z-score**.

5) Pool Metrics: Raw vs Standardized

Recommended default usage: - Pool mean: raw. - Pool minus mean (exclude-self mean): raw. - Pool diff: raw (centered around 0). - Pool rank/percentile: raw (0–1).

Standardize only when: - You need to compare effect sizes across different metric types. - Your model combines variables with incompatible scales.

5A) Rank vs Percentile (when pool sizes vary)

If pool sizes vary greatly, **percentile** is the better default.

Why percentile works better: - Comparable across pool sizes (0.8 always means top 20%). - Stable scaling in pooled models. - Easier to interpret across units and time.

When raw rank can still help: - Pools are uniform in size. - You want absolute position (e.g., “3rd out of 12”). - You are doing within-pool diagnostics only.

Default for this pipeline: - Use **percentile rank** as the standard pool position metric.

6) Partial Effects Plots (PE)

Goal: show hazard ratio vs meaningful x-axis values.

Best practice: - Plot **raw x** even if the model uses z-scores internally. - Evaluate predictions at raw values, then transform to z-scores internally.

For quadratic PE plots: - Plot combined effect: `β1 * x + β2 * x^2`. - Mark turning points if they fall within the observed range.

7) Recommended Default Path

- Use raw ratios and pool means for interpretation.
- Use pool diff for above/below mean signal.
- Use rank/pct for monotonic ordering.
- Add quadratic only when CIF bars show curvature.
- Standardize only after feature construction (squares/interactions).

Quadratic Decision Rule (Quick)

Add a quadratic term when: - CIF bars or CR plots show a U-shape or inverted-U. - Theory suggests effect direction changes at extremes. - Linear vs quadratic model comparison shows better fit.

Do not add a quadratic when: - The relationship is monotonic and stable. - The quadratic flips sign across minor data changes.

Key reminder: Quadratics belong in the **model**, not just the plot. PE plots only show the shape that the model already assumes.

Spline vs Quadratic (Decision Rule)

Use **quadratic** when: - The curve is roughly U-shaped or inverted-U. - You want a simple, stable, easy-to-interpret shape. - You are still validating signal strength and direction.

Use **splines** when: - The curve is clearly non-linear but **not** U-shaped. - You see **plateaus**, **thresholds**, or **multiple bends**. - The quadratic turning point shifts a lot across runs.

Practical workflow: 1) Start with quadratic for your main variables. 2) If the fit is poor or the shape is asymmetric, switch that variable to splines. 3) Keep splines **targeted** to the variables that truly need them.

How to Configure Quadratics (Pipeline Settings)

Quadratics are controlled in config, with base defaults in `pipeline_config.py` and run-specific overrides in a `pip_config_file`.

Simple mode (most common)

Provide a list of base columns to square:

```
QUADRATIC_CONFIG = {
    'enabled': True,
    'bases': [TB_RATIO_SNR_COL, POOL_MINUS_MEAN_SNR_COL],
    'default_zscore': False,
}
```

Advanced mode (full control)

Define explicit terms with names and per-term z-scoring:

```
QUADRATIC_CONFIG = {
    'enabled': True,
    'terms': [
        {'base': POOL_MINUS_MEAN_SNR_COL, 'name': f\"{POOL_MINUS_MEAN_SNR_COL}_sq\", 'zscore': True},
    ],
}
```

Example Guide: Collinearity-Aware Workflow

Use this order when building partial effects plots for tb ratio, pool minus mean, and pool rank in the same run.

Calculation Order

1. Individual TB ratio (raw)
2. Pool mean (raw, inclusive)
3. Pool minus mean (raw, exclude-self mean)
4. Pool diff (raw: `tb_ratio - pool_minus_mean`)
5. Pool rank / percentile (raw, 0–1)

Modeling / Partial Effects Order

Start with single-variable PE plots: - `tb_ratio_*` alone - `pool_minus_mean_*` alone - `pool_rank_pct_*` alone

Then test paired models: - `tb_ratio_* + pool_minus_mean_*` - `tb_ratio_* + pool_rank_pct_*` - `pool_minus_mean_* + pool_rank_pct_*`

Finally, run the full model: - `tb_ratio_* + pool_minus_mean_* + pool_rank_pct_*`

If the full model flattens or flips a shape, treat the PE curves as **conditional** effects and interpret them as “holding the other variables constant.”

Interaction + PE Workflow (Practical)

Use this sequence so PE plots stay informative and interaction plots stay meaningful.

1. **Main effects only**
    - Fit and plot PE curves for each variable alone.
2. **Paired models**
    - Add one additional variable at a time to detect suppression.
3. **Full model with interaction**
    - Add the interaction term and use 3D surface plots for conditional effects.
4. **Interpretation**
    - Main-effect PE plots show unconditional shape.
    - Interaction plots show conditional shape (effect of X depends on Y).

Precompute vs Post-Filter (Time-Saving Rule)

You can precompute most features before Cell 9 filtering.

Precompute before Cell 9: - Raw metrics (tb ratio, pool mean, pool minus mean, pool diff, pool rank/pct) - Quadratic terms (raw → square) - Interaction terms (raw or z, depending on config) - Optional z-scores (if enabled)

Compute after Cell 9: - Model fitting - PE plots and interaction surfaces - Any statistics tied to the filtered sample

Base Default (Raw-First) in the Pipeline

The current default behavior in `pipeline_config.py` is: - **Models use raw variables** (ratios, pool minus-mean, rank/pct). - **Z-scores are computed** (optional) but **not used for model inputs**. - **Interactions are built from raw variables** unless explicitly overridden. - **Plots use raw variables** and exclude z-columns by default.

Run overrides (`pip_config_file`) should only change these when you want a z-score–first or cross-variable comparison run.

Interaction Terms and Collinearity (Why, When, How)

Why interactions: they answer, "Does the effect of X depend on Y?"
Example: the effect of `tb_ratio` may be stronger or weaker depending on `pool_minus_mean` or `pool_rank_pct`.

How interactions create collinearity: an interaction term is a product of two variables that are often correlated with the originals. This can make coefficients unstable and cause the **main-effect PE curves** to flatten or flip.

How to interpret PE plots with interactions: - Main-effect PE curves become **conditional** (effect of X when Y is held fixed). - The 3D surface plot is the **joint effect**, which often carries the shape.

Suggested modeling order: 1) Fit and plot single-variable PE curves (no interactions). 2) Add quadratics if CIF/CR suggests curvature. 3) Add the interaction and inspect the 3D surface. 4) Expect main-effect curves to compress once the interaction is included.

Precompute vs Post-Filter (What to Do Early)

You can safely precompute **all feature construction** before Cell 9: - Raw metrics (tb ratios, pool means, pool minus mean, pool diff, ranks/pct) - Quadratic terms (built from raw columns) - Interaction terms (built from raw columns) - Z-scores (if you want them available but not necessarily used)

What must happen **after Cell 9 filtering**: - Model fitting (coefficients are sensitive to the filtered sample) - PE plots and interaction surfaces (these are model-dependent) - Any model diagnostics or significance tests

This keeps the pipeline fast and repeatable: heavy feature creation is done once, and all modeling steps are re-run only on the filtered dataset.